Investing in the Long Run

Chris Mulligan

Class of 2025

Mathematics and Economics

Nicholas Cifelli

Class of 2028

Mathematics and Economics

Griffin Mulcahy

Class of 2025

Mathematics and Physics

Jack Perdue

Class of 2026

Mathematics and Architecture

Mathematical Methods in Financial Economics (MATH 40570)

University of Notre Dame

May 2025

Contents

1 Introduction **4**

2 Basic Financial Instruments **4**

2.1 Cash 4

2.2 Savings Accounts 5

2.3 Bonds 5

2.4 Stocks 6

2.5 Mutual Funds 7

2.6 Exchange Traded Funds 7

2.7 Individual Retirement Accounts 8

2.8 401(k) 9

3 Traditional Methods of Investing and Constructing Portfolios **10**

3.1 Modeling Assets to Improve Portfolio Construction 13

4 State of AI in Academia and Finance Industry and the Outlook of Both **16**

5 Limitations of AI in Finance **18**

6 Using AI to construct a long-term portfolio **19**

6.1 Question One: What average yearly return and variance should I strive for? 19

6.1.1 ChatGPT 19

6.1.2 Grok 20

6.1.3 Gemini 20

6.1.4 Perplexity 20
6.1.5 Copilot 21
6.1.6 Table of targeted returns and volatilities 21
6.2 Question Two: What types of assets should I include in my long-run investing portfolio? 22
6.2.1 ChatGPT 22
6.2.2 Grok 23
6.2.3 Gemini 24
6.2.4 Perplexity 24
6.2.5 Copilot 24

7 Designing a Long-Run Portfolio **25**
7.1 ETF Portfolio 27
7.2 Mutual Fund Portfolio 29

8 Conclusion **33**

9 References **34**

10 Code Appendix **37**
10.1 Code for Modeling Single Stock 37
10.2 Code for confidence intervals 41
10.3 60/40 portfolio code 47
10.4 Efficient Frontier Portfolio Design 56

11 AI Appendix **65**
11.1 Question 1: What average yearly return and variance should I strive for in my long-run investing portfolio over a 20 year period? 65

11.2 Question 2: What types of assets should I include in my long-run investing portfolio? 68

1 Introduction

Long-term investments are investments not planned to be liquidated in the near future,investments typically planned to last for at least 10 years. Of all types of investment, long-term investment is the most common and useful form for the general public. This paper discusses the most basic and common forms of investment options available to most investors as well as traditional methods of constructing portfolios available to an average person. The paper explores implications of the pressing issue of AI, which is poised to have a seismic impact on the world of investing, considering a review of the current uses and roles of AI in investment, then speculating at potential future uses in the context of recognized limitations inherent to AI and machine learning.

2 Basic Financial Instruments

We will first consider basic financial instruments available to investors to construct their portfolios for the long term.

2.1 Cash

The most straightforward investment one can make is simply holding their assets in cash. Cash is fully liquid, which means that it is the easiest form to exchange for goods and services, and other investments are exchanged for cash. While it is generally advised to keep some wealth in cash for short-term transactions, holding all of your assets in cash is generally ill advised due to inflation. Since the supply of cash is always increasing, cash loses it's purchasing power, which is what cash can be physically exchanged for, over time. Additionally, since cash provides no

return, it normally makes sense to invest in one of the other investments that will be introduced shortly, which generally provide a return for long-run investing.

2.2 Savings Accounts

One common way that investors gain a safe return on their assets is by putting their cash into a savings account in a bank. This investment is very safe in the modern era as the only risk is if the bank which you create a savings account with goes under. And, even if the bank collapses, your account will likely be insured by the FDIC meaning that the government protects much of the money in savings accounts in the event of bank collapses. However, while these assets are incredibly safe, they generally provide a very minimal return. Currently, the average savings account APY in the USA is 0.59. That being said, recently the option of high yield savings accounts, some offering APY of up to 5% exist. (All data came from https://www.bankrate.com/banking/savings/average-savings-interest-rates/)

2.3 Bonds

Bonds are an investment vehicle that essentially constitute a loan to another organization that is to be repaid at a future date with interest. At the transaction, the buyer pays the principal on a bond and over time the get paid back with interest, either once the bond reaches maturity, through periodic coupons, or a mix of the two. Some bonds, like government debt bonds, are practically risk free. However, non-government bonds, those issued by corporations or other organizations, come in varying risk levels, and are rated by by S&P on a scale from AAA (highly safe) all the way down to D (commonly referred to as junk bonds). Generally, as the rating gets lower, the interest is higher since the investor assumes a higher risk

that they might never be repaid. Some examples of popular bonds include US government bonds, corporate bonds, foreign government bonds, federal agency bonds, asset-backed securities (a popular one is the mortgage-backed security), and local government bonds.

2.4 Stocks

Stocks represent partial ownership of a public company. Stocks are traded on centralized exchanges such as the NASDAQ or the NYSE in the US or foreign exchanges such as the London SE, Euronext Paris, or the Japanase TYO. The perks of stock ownership vary from company to company but may include granting an investor the right to vote at shareholder meetings or receiving a share of company profits in the form of dividends among other things. However, some stocks do none of these things, and in these cases, the stock owner can only benefit from the sale of stocks after a price rise or from other trading strategies such as short selling. Historical data has verified that there exists a risk equity premium when comparing stocks and bonds. Over time, this risk equity premium has been calculated to be 7.64%, meaning that the average annual return on stocks is 7.64% greater than the average annual return on safe government bonds. Stocks are very valuable because of this fact and are, as such, necessary in an ideal long-term portfolio, however, they are generally much more volatile than bonds. Usually, it is not advised to put all of one's assets into a single stock but rather to diversify and spread investment over the entire market so that events such as the bankruptcy of one company do not completely tank a portfolio.

2.5 Mutual Funds

A mutual fund is an investment vehicle where a group of investors pool a large amount of assets which are then professionally managed. The managers of a mutual fund usually invest in a highly diversified portfolio consisting of many different type of stocks and bonds. Mutual funds allow average investors to get a share of large-scale investments that they would not be able to manage or monitor on their own. The drawback of this is that there are fees associated with mutual funds that do not exist in direct investment because of the need to pay the middle man. By investing in mutual funds, an investor pays a premium to avoid the risks of direct investment and to have the benefit of someone constantly monitoring and adjusting their portfolio. As such, mutual funds are useful for average investors who wish to gain diversified exposure to more categories of stocks and bonds.

2.6 Exchange Traded Funds

An ETF is an investment vehicle that allows an investor to buy and sell shares in a single security that represents a fractional ownership of a portfolio of securities. For example, ETFs can represent a stock index, such as the $SPY ETF which represents an investment in the S&P 500. Or, ETFs can also represent a single sector such as the $VGT ETF which represents an investment in the technology sector. Overall, ETFs are useful tools when an investor wants to diversify in a specific area of the market but doesn't have the time to individually interact with hundreds of different securties.

2.7 Individual Retirement Accounts

An Individual Retirement Account, commonly referred to as an IRA, is an investment vehicle designed to help working individuals save for their retirement in a structured and efficient way that minimizes the taxation of their retirement funds. Rather than being investment strategies themselves, IRAs act as investment containers capable of holding stocks, bonds, ETFs, or other assets. There are two primary types of IRAs: traditional and Roth. A Traditional IRA allows contributions to grow tax-deferred: taxes are paid only upon withdrawal, often during retirement when one's income may be lower. Traditional IRAs allow the value of the taxed income to compound before being assessed. In contrast, a Roth IRA is funded with assets already taxed, but offers the significant advantage of tax-free growth and tax-free withdrawals in retirement. Both types of IRAs allow investors to minimize the consequences of the taxation of their income.

Designed purely as a long-term savings tool, IRAs penalize early withdrawals and limit annual contributions. For 2024, account holders were limited to saving $7,000 (or $8,000 for individuals fifty years and older), encouraging long-term saving habits such as preparing for retirement. Though primarily in the form of personal accounts, IRAs may also be employer-based, such as SEP and SIMPLE IRAs which cater to self-employed individuals and small businesses. In general, IRAs serve as a powerful tool in retirement planning, offering flexibility, investment control, and valuable tax benefits, making them a useful tool to promote the secure financial futures of those who hold them.

2.8 401(k)

A 401(k) is a retirement savings plan sponsored by an employer. It allows employees to save and invest a portion of their paycheck before taxes are taken out. Often, employers will also contribute to the plan, sometimes matching a percentage of the employee's contributions. These plans offer tax advantages, as the money grows tax-deferred until retirement, and in some cases (Roth 401(k)), qualified withdrawals in retirement are tax-free.

3 Traditional Methods of Investing and Constructing Portfolios

Portfolio construction and traditional methods of investing have long served as the foundation for financial planning. Core principles such as diversification, asset allocation, and risk management have long been essential to evaluating and constructing portfolios. Construction strategies consider investor's financial goals and risk tolerance, aiming to balance potential returns with acceptable levels of risk.

A central aspect of portfolio construction, asset allocation is the process of distributing investments across various asset classes, like stocks, bonds, and cash equivalents, to optimize the balance between risk and return. In J.P. Morgan's informational guide on portfolio structuring, (J.P. Morgan Asset Management, 2021), asset allocation constitutes a primary role of portfolio structuring. A traditional and widely accepted asset allocation approach is the 60/40 portfolio. In this approach, an investor allocates 60 percent of his or her assets to equities and 40 percent to fixed-income securities. According to (Sivaraman, 2023), historically, this mix has provided consistently-desirable long-term returns, though ongoing evaluation continues to promote other contrasting allocation strategies.

Critical to the stability of a portfolio and the minimization of its volatility, diversification serves a critical role as both a metric of a portfolio's strength and a tool through which portfolios may be effectively constructed. As supported by (Investopedia, (n.d.)), by investing in a broad range of assets within each class, diversification seeks to reduce the impact of any single investment's performance on the overall

portfolio. This strategy enhances resilience against market volatility.

Diversification may be measured in multiple distinct ways. It is important to consider and limit covariance between elements of a portfolio, and diversification makes it possible to limit this type of portfolio limitation. Diversification across various countries and market sectors contributes to the overall stability of a portfolio as well.

One of the primary means of tailoring a portfolio to a particular investor is through risk assessment. Understanding personal risk tolerance is crucial in shaping an investment strategy. When constructing a portfolio, it is important to consider and assess a utility function for a particular investor and understand his or her particular level of aversion to risk. Aligning a portfolio's risk profile with an investor's comfort level and financial goals helps in maintaining discipline during market fluctuations as well as meeting a particular investor's preferences.

Companies specializing in the construction of client investment portfolios often assess risk through questionnaires or research on client attitudes and preferences, such as the quiz offered by Vanguard:
https://personal.vanguard.com/us/FundsInvQuestionnaire.

The constant evolution of investment strategies has been extensively documented. From (Emerald, 2023), a comprehensive literature review covering the period from 1900 to 2022 traces the development of stock market investment approaches, highlighting shifts from traditional methods to modern techniques influenced by technological advancements. Additionally, the CFA Institute offers insights into both fundamental and systematic investment strategies, reflecting the continuous adap-

tation of portfolio construction practices to meet changing market conditions. As supported by Novus (Altshuller, 2017), though much of this information is publicly available, data pertaining to historical portfolio performances and covariances of stocks is often coveted and held private by institutions seeking to gain advantage over the general public. In fact, according to a study by Novus of diverse portfolio performances over time, (Chadwick, 2022), an overwhelming majority of eighty percent of hedge fund managers do not publish their full portfolios with investors.

In analyzing their studies, Novus (Altshuller, 2017) finds that managers of hedge funds and other investment portfolios are reluctant to share their data and information. Out of caution against the possibility that their processes could be replicated or reconstructed from their data, managers actively avoid the possibility that data pertaining to their positions is passed to competitors. This desire for privacy over each investment manager's data stimulates a larger culture of secrecy that opposes transparency for public access to data on various portfolio and investment performances.

While traditional strategies have proven effective over time, contemporary market environments present new challenges. For instance, according to (Sivaraman, M., 2023), the classic 60/40 portfolio has recently shown signs of strain, prompting investors to consider diversifying beyond conventional asset classes. Furthermore, the rise of alternative investment vehicles and strategies, such as private equity and hedge funds, offers additional avenues for diversification and potential returns. However, according to (Smith, 2025) of Financial Times, these alternatives often come with increased complexity and risk. For example, in March of 2025, the stocks of larger private equity and asset management houses were affected much more nega-

tively during the market correction than other portfolios.

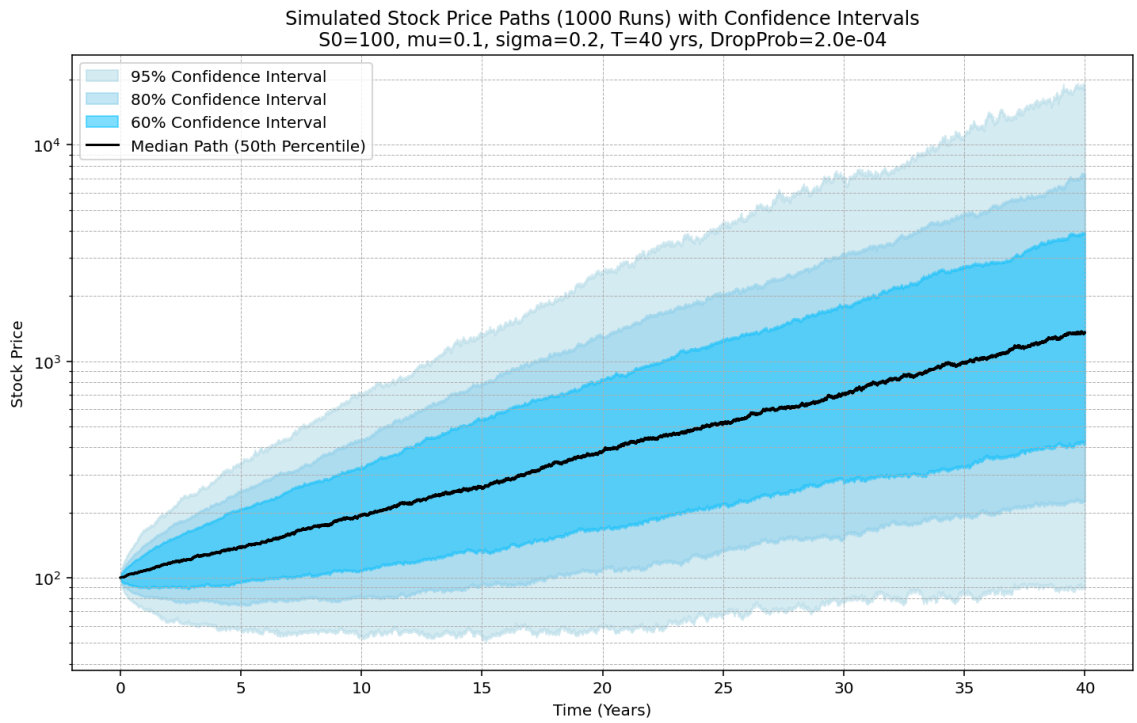
Traditional investment methods and portfolio construction techniques remain foundational in financial planning. However, staying informed about evolving strategies and market dynamics is essential for optimizing investment outcomes in today's complex financial landscape.

3.1 Modeling Assets to Improve Portfolio Construction

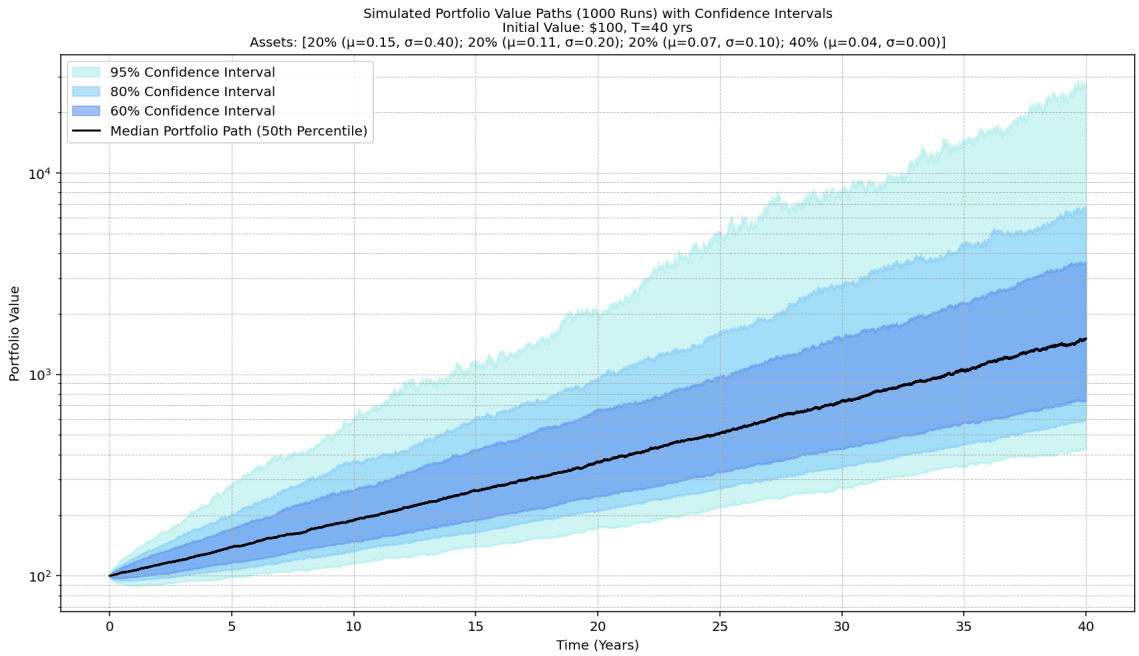
An important factor of constructing long term portfolios is being able to successfully model how certain assets will tend to move in the long run. The most important of these assets to model are stocks, since out of the long term options they attain the best returns and have the highest variance. A normal way to model stocks is by having them follow the stochastic differential equation

$$dS_t = aS_t dt + \sigma S_t W_t, \;\; t \geqslant 0$$

Where S_t is the price of the stock at time t, a is the return of the stock, σ is it's standard deviation, and W_t is a Wiener process. This model is known to do well at predicting stock prices in the short to medium term. The issue with using the model over the long run is that the model does not account for random large market downturns which are known as recessions. Recessions happen about every 8 years on average and can be drops where the market loses anywhere from 10-25+ percent. Thus we created a model, (see code appendix for full details) that factors in these random recessions and gives us a better picture of long run stock movement. We created this model, then plotted the results with 60,80, and 95 percent confidence intervals.



This visual demonstrates how even with random recessions the median stock will make very good returns, more than 10x, over 40 years. It also underscores a fundamental point in portfolio construction which is the extreme importance of diversification. As this graph shows, even over 40 years, due to the volatile nature of stock, there is a chance that your stock loses value, or fails to outpace inflation. However, if you invest in a large basket of stocks, then you will generally make a very good return. Overall, this model that we developed to model long term stock prices with recessions provides useful insight into the nature of long term investments and supports the traditional methods of portfolio creation. The only issue with this model, of only investing in stocks, is that the 80% confidence interval spans from 2x - 70x. Thus we simulate a traditional 60/40 portfolio, with the stocks split evenly between small, mid, and large caps. Here is the simulation



Taking into consideration that a bond with return 0.04 would grow 4.8x over 40 years, we see that this portfolio clearly beats that handily since 90 percent of these portfolios will return more than 6x. Thus, the traditional method of portfolio construction is clearly useful. Modeling data shows that the 60/40 split consistently beats pure bond returns with very high probability. This is very useful for an investor who wants to set it and forget it. However, as hedge funds have done in the past, constant portfolio monitoring and readjustment can optimize these returns greatly.

4 State of AI in Academia and Finance Industry and the Outlook of Both

There are numerous current applications of AI and ML (AIML) in investing, and AIML has the potential to fundamentally change investment strategy and portfolio management through advanced processing of (non-traditional) data, predictive analytics, and automation. Going forward, financial firms seek to leverage these tools to enhance decision making, optimize risk management, and, in the short term, as discussed above, hopefully gain a competitive advantage.

AI-driven algorithms analyze real-time market data as well as judge investor sentiment through social media and the news, and combine these insights with historical data to dynamically adjust asset allocation to satisfy an investor's risk tolerance.

AI is also being employed to help support passive index-based investing strategies to ensure portfolios track their benchmark indices as closely as possible.

On the active management side, reinforcement learning is being used to enable millisecond trading decisions and make adjustments.

In addition, AIML provides enhanced risk assessment and fraud prevention capabilities. AIML can use reinforcement learning to detect anomalies in transaction patterns and identify fraud in real time.

Generative AI uncovers undervalued assets and emergent market trends by analyzing vast data sets, including non-traditional sources like earnings call transcripts.

Finally AIML can improve current robo-advisors by tailoring recommendations very closely to an individual's risk tolearance and financial goals.

According to (Goodell, J.; et al. 2021) the use of AI in finance is not limited to identifying arbitrage opportunities and can be divided into three categories. (1)

portfolio construction, valuation, and investor behavior; (2) financial fraud and distress; and (3) sentiment inference, forecasting, and planning. Other areas of interest involve portfolio selection and risk management where, presently, AI complements human analysts at some firms (Antaki, M. 2025). But it is predicted that over the next 3-5 years AI will be able to make more risk assessments and similar analysis tasks with less supervision, leading to a restructuring of quantitative trading teams (Antaki, M. 2025).

According to The Guardian, The Bank of England's Financial Policy Committee has warned that, "(AIML) models might learn that stress events (periods of extreme volatility) increase their opportunity to make profit and so take actions actively to increase the likelihood of such events" ands the Guardian reports that, "Those AI programs may 'identify and exploit weaknesses' of other trading firms in a way that triggers or amplifies big moves in bond prices or stock markets" to facilitate profit opportunities. Ultimately, the Guardian reports, "the models could 'facilitate collusion or other forms of market manipulation ... without the human manger's intention or awareness' " (Markortoff, K. The Guardian 2025).

5 Limitations of AI in Finance

When an investment firm chooses to invest in AI, that is because they are fundamentally rejecting the Efficient Market Hypothesis (EMH). That is because the reason they are choosing to use AI is because they believe that they can use the AI to gather and interpret information about an asset before the market adjusts that will give them an advantage over other investors. If the firms believed the EMH, then this would be nonsensical since gathering this information would not be helpful since the information was already pre-factored into the pricing of the asset. Thus, firms are investing in AI because they believe that they can exploit market inefficiencies before others because of the data processing advantage that AI provides.

However, this causes an inherent tension. If AI agents become widespread throughout the investment world, then the market itself will become more efficient. Thus, AI will diminish the market inefficiencies that it was created to exploit, limiting it's usefulness. As such, the logical scenario is that the select firms will keep their AI tools safe, so that they can continue to exploit market inefficiencies that exist while not simultaneously destroying said inefficiencies. This is already playing out in real time, large investment companies are not releasing their AI agents to the public so that the advantage stays with them. This will further extend the gap between an everyday investor and the large firms.

Beyond the EMH, there are other issues inherent to prediction that arise when considering the limits on AI. One of these is the phenomenon of a reflexive prediction. This is when knowledge/disclosure of a prediction can cause an alteration in the way other investors in the market react, fundamentally altering the situation from what

was predicted to happen. AI cannot escape this boundary, it is simply a part of behavioral economics. Another limit is so called black swan events. These are events that are incredible rare and highly impactful. An example of a black swan event could be a natural disaster, a terrorist attack, or recently as with United Health Group, the murder of an executive, all of which could be highly impactful to the price of any asset but are also realistically unpredictable.

6 Using AI to construct a long-term portfolio

6.1 Question One: What average yearly return and variance should I strive for?

To consider the potential advantages and effects of AI in long-term investment, we have compiled responses across multiple leading AI software platforms to find what current models suggest as priorities in investment and what strategies each promote to employ. We choose to examine responses from ChatGPT, Grok, Gemini, Perplexity, and Copilot. We prompted each model with the following question: "What average yearly return and variance should I strive for in my long-run investing portfolio over a 20 year period?"

6.1.1 ChatGPT

ChatGPT discussed three different portfolio models corresponding to differing levels of aggression and risk. The portfolios considered were for a "Moderate to Aggressive Portfolio with 60-80% equities;" a "Moderate risk portfolio with 60% allocation to stocks and 40% to bonds;" and an "Aggressive portfolio with 80–100% allocation to stocks." ChatGPT analyzed and provided expected real and nominal

returns as compared with historical SP 500 returns and the expected volatility associated with these portfolios. ChatGPT concluded that an investor should target an annual return of 7% nominal or 5% real return with a target variance of 0.01–0.02. Additionally the Sharpe Ratio finds that greater than 0.5 is solid and 0.7–1.0+ is excellent.

6.1.2 Grok

Grok stated that an investor should target an average yearly return of 6-8% nominal or 4-6% real annually. It found that with this targeted return, an acceptable variance would fluctuate between -10% and +15%, with occasional worse years during market downturns, pointing out that over 20 years, the compounding effect should smooth out volatility.

6.1.3 Gemini

Much like ChatGPT, Gemini discussed three different portfolio models corresponding to differing levels of aggression and risk. The portfolios considered were for an "Aggressive Investor with High Risk Tolerance;" a "Moderate Investor with Medium Risk Tolerance;" and a "Conservative Investor with Low Risk Tolerance." Gemini discussed the importance of accurately determining an investor's conformability with volatility and determining a utility function corresponding to a particular investor's goals and tolerances.

6.1.4 Perplexity

Similarly, Perplexity discussed three different portfolio models corresponding to differing levels of aggression and risk, suggesting to base allocations on risk tolerance,

always reinvest dividends, and re-balance portfolios annually. Perplexity concluded that for most investors, a 7-9% inflation-adjusted return with annual volatility under 15% strikes a prudent balance between growth and risk over two decades and is achievable through a diversified portfolio.

6.1.5 Copilot

Copilot gave similar responses prioritizing the importance of tailoring a portfolio to the particular goals and tolerances of the investor. Copilot considered two primary types of portfolios targeting returns around 7-10% or 10-15% with their corresponding volatility.

6.1.6 Table of targeted returns and volatilities

AI Model	Allocation	Target Return	Hist. Return	Std. Dev.	Variance
ChatGPT	80–100% stocks	-	-	˜15–20%	-
Grok	-	-	-	-	-
Gemini	mostly stocks	8–10%+	8–10%+	15%+	-
Perplexity	80–100% stocks	7.6–10.4%	-	-	-
Copilot	-	7–10%	-	15–20%	0.0225 and 0.04

Table 1: Aggressive Portfolio by AI Model

AI Model	Allocation	Target Return	Hist. Return	Std. Dev.	Variance
ChatGPT	60–80% equities	˜8–10%	˜6–8%	-	-
Grok	-	-	-	-	-
Gemini	60/40 split	7–9%	10–14%	-	-
Perplexity	60% stocks / 40% bonds	˜8.8%	-	-	-
Copilot	-	6–8%	10–15%	-	0.01–0.0225

Table 2: Moderate Portfolio by AI Model

AI Model	Allocation	Target Return	Std. Dev.
ChatGPT	60% stocks / 40% bonds	10–12%	5% — 7%
Grok	-	4–6% — 6–8%	-
Gemini	-	4–6%	¡10%
Perplexity	-	7–9%	¡15%
Copilot	-	-	-

Table 3: Conservative Portfolio by AI Model

6.2 Question Two: What types of assets should I include in my long-run investing portfolio?

6.2.1 ChatGPT

ChatGPT's response describes the different asset classes and discusses recommended allocations to each: Equities (target allocation: 60-80% based on risk tolerance), Bonds (target allocation: 10-30% choosing duration based on risk tolerance and time horizon), Physical assets (including real estate, real estate investment trusts, commodities like oil, gold, and agriculture that hedge against inflation), and alternative investments (including crypto, venture capital, and hedge funds).

AI Model	Target Return	Variance	Volatility
ChatGPT	5% — 7%	0.01–0.02	-
Grok	4–6% — 6–8%	-	-10% – +15%
Gemini	-	-	-
Perplexity	-	-	-
Copilot	-	-	-

Table 4: Target Portfolio by AI Model

ChatGPT also gave tips for portfolio construction: diversify across countries and markets, use low-fee index funds and ETFs to maximize compounding, rebalance regularly, at a minimum yearly, to keep risk in check. Dollar-cost averaging to minimize the impact of volatility, stay invested in the market, it quotes the common phrase "Time in the market beats timing the market".

6.2.2 Grok

Grok recommends constructing a long-run portfolio around six complementary pillars: equities to drive growth, fixed income to smooth volatility, real estate as an inflation hedge and income source, cash and equivalents for liquidity, alternative assets to broaden diversification, and tax-advantaged retirement accounts to maximize compounding. Contextualizing the strategy for 2025 includes attention to stabilizing interest rates, continued strength in technology and renewable sectors, and growing emphasis on ESG factors. A moderate-risk illustration allocates roughly 60% to equities, 20% to bonds, 10% to real estate, 5% to cash, and 5% to alternatives, with guidance to favor low-cost index funds, rebalance annually, dollar-cost average, and remain fully invested through market cycles.

6.2.3 Gemini

Gemini lays out a classic framework in which domestic and international equities—including large-, mid-, and small-cap stocks—serve as the primary engine for long-term growth, while fixed income instruments such as U.S. Treasuries, TIPS, and corporate or municipal bonds provide income and risk mitigation. Real estate exposure via REITs or direct property ownership offers inflation protection, and modest allocations to commodities and cash equivalents further enhance diversification and liquidity. The "core-and-satellite" approach anchors the portfolio in broad, low-cost index funds as the core allocation, complemented by targeted satellite positions, with periodic rebalancing aligned to an investor's time horizon and risk tolerance.

6.2.4 Perplexity

Perplexity condenses the strategy into four narrative threads: stocks across various market capitalizations and geographies act as the primary growth engine; bonds ranging from government to high-yield deliver income and stability; cash and equivalents preserve liquidity during downturns; and supplementary assets like REITs and commodities (notably gold) provide inflation hedging and low-correlation ballast. Illustrative allocations span an aggressive near-100% equity stance, a balanced 60/40 equity-to-fixed income mix, and a conservative bond-plus-cash tilt, all within a framework that emphasizes low-fee index products and regular portfolio reviews.

6.2.5 Copilot

Copilot emphasizes resilience through broad diversification across equities, fixed income, real estate, commodities, alternative investments, and cash equivalents. A "core-and-satellite" construction positions diversified, low-cost ETFs or mutual funds

spanning stocks and bonds at the portfolio's core, while satellite allocations—such as real estate or niche alternatives—add targeted exposure. Over a 20-year horizon, disciplined rebalancing is highlighted as essential to maintain target allocations and ensure that portfolio risk remains aligned with long-term objectives.

7 Designing a Long-Run Portfolio

We will now use historical data to build a long-run portfolio. To start, if we take the median retirement age of 62 and the median age of a college graduate of 22, we see that investors typically spend about 40 years in the workforce. Thus, by just investing in the risk-free rate of 4.38% per year, investors will earn 456% after 40 years. However, investors can increase their return significantly by taking on some risk.

Table 5: **Total return by yearly return and length of time**

	1 Year	2 Years	5 Years	10 Years	15 Years	20 Years	25 Years	30 Years	35 Years	40 Years
1.0%	1.0	2.0	5.1	10.5	16.1	22.0	28.2	34.8	41.7	48.9
2.5%	2.5	5.1	13.1	28.0	44.8	63.9	85.4	109.8	137.3	168.5
4.4%	4.4	9.0	23.9	53.5	90.2	135.7	192.0	261.8	348.3	455.5
5.0%	5.0	10.3	27.6	62.9	107.9	165.3	238.6	332.2	451.6	604.0
7.5%	7.5	15.6	43.6	106.1	195.9	324.8	509.8	775.5	1,156.9	1,704.4
10.0%	10.0	21.0	61.1	159.4	317.7	572.7	983.5	1,644.9	2,710.2	4,425.9
12.5%	12.5	26.6	80.2	224.7	485.2	954.5	1,800.3	3,324.3	6,070.8	11,019.9
15.0%	15.0	32.3	101.1	304.6	713.7	1,536.7	3,191.9	6,521.2	13,217.6	26,686.4
17.5%	17.5	38.1	124.0	401.6	1,023.5	2,416.3	5,535.7	12,522.2	28,170.0	63,216.2
20.0%	20.0	44.0	148.8	519.2	1,440.7	3,733.8	9,439.6	23,637.6	58,966.8	146,877.2

To build a safe long-run portfolio, we examined the returns on the biggest ETFs and Mutual Funds by market sector.

Table 6: **Biggest ETF and Mutual Funds by Market Sector**

Sector	Biggest ETF	Biggest Mutual Fund
Technology	VGT	VITAX
Financial Services	XLF	VFAIX
Consumer Cyclical	XLY	BPTRX
Healthcare	XLV	VGHCX
Communication Services	VOX	VTCAX
Industrials	XLI	VINAX
Consumer Defensive	XLP	VCSAX
Energy	XLE	VENAX
Real Estate	VNQ	VGSLX
Basic Materials	XLB	VMIAX
Utilities	XLU	VUIAX

We took the daily, weekly, and monthly returns for each of these assets over 1 year, 2 year, 5 year, 10 year, 15 year, and 20 year horizons to get a good look at how the assets interact with each other. We decided that 15 years would be a long enough horizon to secure good long-term data without being affected by the 2008 Great Recession. While recessions do occur, the 2008 Great Recession was an outlier and should not affect our historical data. Additionally, we decided that, for a long-run portfolio, taking monthly returns is optimal. This way, we are unaffected by the variances and covariances of daily or weekly returns when our only focus is long-term growth.

7.1 ETF Portfolio

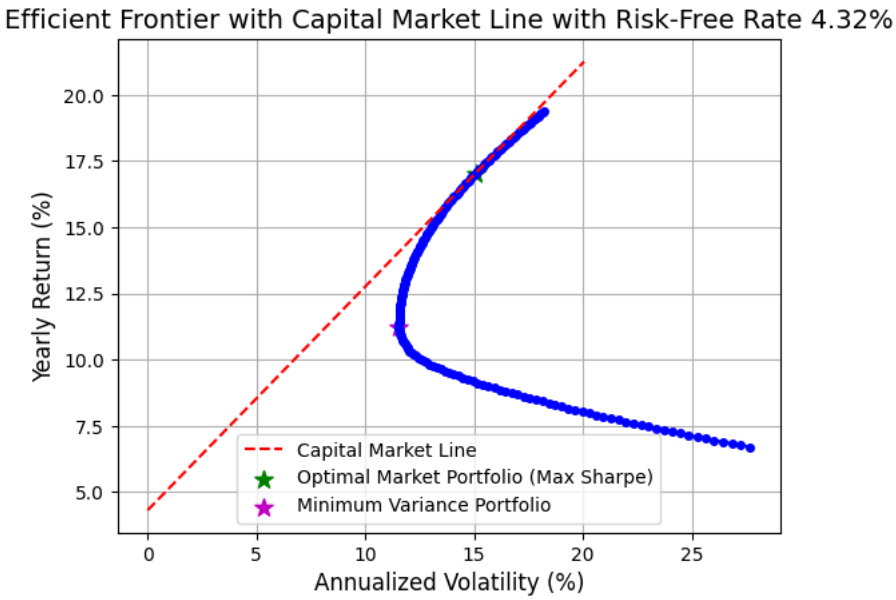
First, we found the past 15 year monthly historical data of the ETFs in our portfolio:

Table 7: **15 Year Monthly Historical Performance of ETF Securities**

Ticker	Sector	Total Return	Yearly Return	Volatility	Annualized Volatility
NASDAQ	Market	814.03%	15.99%	4.92%	17.06%
VGT	Technology	1,306.78%	19.39%	5.27%	18.24%
XLF	Financial Services	485.08%	12.57%	5.63%	19.49%
XLY	Consumer Cyclical	868.34%	16.44%	5.35%	18.52%
XLV	Healthcare	505.62%	12.83%	3.92%	13.59%
VOX	Communication Services	339.03%	10.43%	4.77%	16.52%
XLI	Industrials	572.78%	13.63%	5.27%	18.26%
XLP	Consumer Defensive	351.12%	10.63%	3.55%	12.29%
XLE	Energy	163.26%	6.70%	7.99%	27.68%
VNQ	Real Estate	285.74%	9.47%	5.29%	18.33%
XLB	Basic Materials	307.06%	9.87%	5.64%	19.52%
XLU	Utilities	345.29%	10.53%	4.27%	14.80%

Using that data, we found the optimal weights to build a variance-minimizing portfolio that achieved a specific yearly return for all possible yearly returns between the minimum and maximum above:

Figure 1: **Efficient Frontier of ETFs**



To summarize, we provide the weights to achieve the minimum variance at select points on the efficient frontier, in addition to the weights for the optimal portfolio with the highest Sharpe ratio and the minimum variance portfolio:

Table 8: **Optimal Portfolio for ETF Securities at Various Returns**

		Yearly Return				Optimal Portfolio	Minimum Variance Portfolio
Ticker	Sector	7.5%	10.0%	12.5%	15.0%	17.0%	11.2%
VGT	Technology	0.00%	0.00%	14.20%	43.36%	69.87%	0.00%
XLF	Financial Services	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
XLY	Consumer Cyclical	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
XLV	Healthcare	0.00%	0.00%	29.45%	27.00%	13.50%	28.91%
VOX	Communication Services	0.00%	5.67%	0.00%	0.00%	0.00%	6.52%
XLI	Industrials	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
XLP	Consumer Defensive	0.00%	53.69%	32.89%	4.48%	0.00%	42.84%
XLE	Energy	73.77%	15.09%	0.00%	0.00%	0.00%	0.00%
VNQ	Real Estate	19.70%	0.00%	0.00%	0.00%	0.00%	0.00%
XLB	Basic Materials	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
XLU	Utilities	6.53%	25.55%	23.46%	25.16%	16.63%	21.72%
Annualized Volatility		22.86%	12.58%	11.69%	13.04%	15.06%	11.55%

7.2 Mutual Fund Portfolio

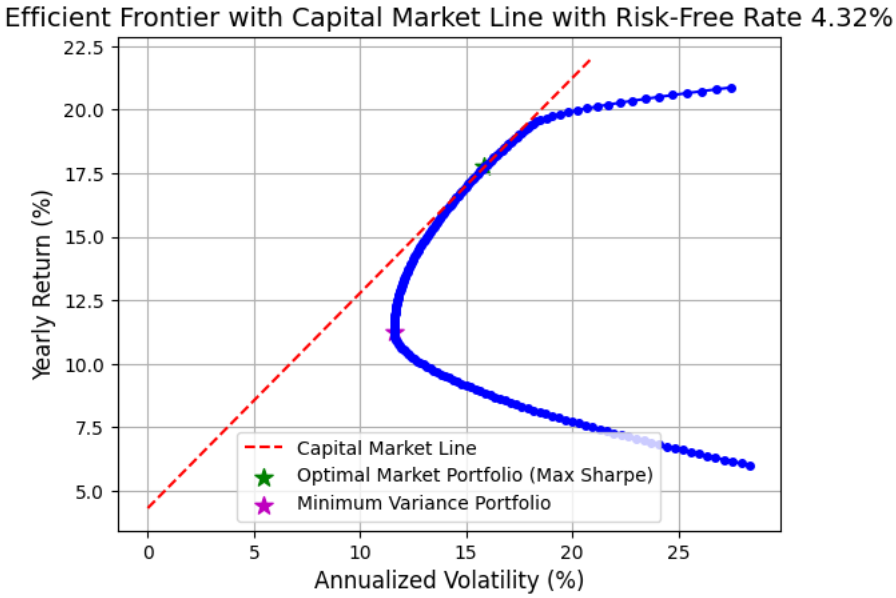
Next, we will conduct the same work for the biggest mutual funds in each market sector. First, we see the following 15-year monthly returns data:

Table 9: **15 Year Monthly Historical Performance of Mutual Fund Assets**

Ticker	Sector	Total Return	Yearly Return	Volatility	Annualized Volatility
NASDAQ	Market	814.03%	15.99%	4.92%	17.06%
VITAX	Technology	1,322.64%	19.48%	5.25%	18.18%
VFAIX	Financial Services	493.72%	12.68%	5.66%	19.59%
BPTRX	Consumer Cyclical	1,590.83%	20.87%	7.93%	27.49%
VGHCX	Healthcare	432.86%	11.87%	4.67%	16.17%
VTCAX	Communication Services	351.55%	10.64%	4.74%	16.43%
VINAX	Industrials	574.08%	13.65%	5.38%	18.63%
VCSAX	Consumer Defensive	380.05%	11.09%	3.51%	12.15%
VENAX	Energy	138.76%	6.01%	8.19%	28.38%
VGSLX	Real Estate	284.67%	9.45%	5.27%	18.27%
VMIAX	Basic Materials	322.28%	10.14%	5.82%	20.16%
VUIAX	Utilities	348.85%	10.59%	4.21%	14.58%

From that data and performing variance minimization for several levels of return, we see the following efficient frontier and capital market line:

Figure 2: **Efficient Frontier of Mutual Fund Securities**



Now, we gather the weights at various points on the efficient frontier in addition to both the optimal portfolio and the minimum variance portfolio:

Table 10: **Optimal Portfolio for Mutual Fund Securities at Various Returns**

		Yearly Return				Optimal Portfolio	Minimum Variance Portfolio
Ticker	Sector	7.5%	10.0%	12.5%	15.0%	17.7%	11.2%
VITAX	Technology	0.00%	0.00%	16.94%	47.27%	80.36%	1.88%
VFAIX	Financial Services	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
BPTRX	Consumer Cyclical	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
VGHCX	Healthcare	0.00%	0.00%	13.76%	7.59%	0.00%	15.99%
VTCAX	Communication Services	0.00%	3.00%	0.00%	0.00%	0.00%	3.96%
VINAX	Industrials	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
VCSAX	Consumer Defensive	0.00%	43.91%	45.45%	21.75%	0.00%	54.68%
VENAX	Energy	63.39%	17.71%	0.00%	0.00%	0.00%	0.00%
VGSLX	Real Estate	16.32%	0.00%	0.00%	0.00%	0.00%	0.00%
VMIAX	Basic Materials	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
VUIAX	Utilities	20.29%	35.39%	23.86%	23.39%	19.64%	23.49%
Annualized Volatility		20.96%	12.93%	11.77%	13.13%	15.85%	11.61%

Hence, readers can now set themselves up with a long-run portfolio of either mutual funds or ETFs and the risk-free rate. It is recommended that the reader re-run this code every year, using a rolling 15-year estimation window, to rebalance their portfolio.

8 Conclusion

Overall, this project has been incredibly useful for us to undertake as young investors who will soon have to begin constructing our own long-term portfolios. We have explored and become familiar with the investment vehicles that are commonly used in long-term portfolios. We have explored and modeled traditional methods of portfolio construction. We have been able to research and discuss AI and it's current uses in this field as well as it's future and it's inherent limitations. To conclude, we constructed two long term portfolios using the Markowitz method, one for ETFs and one for Mutual Funds. This construction was very useful and served as a great exercise for us. In summary, this project has been rewarding four our group, and has provided us with great insights to serve us well in our future investment endeavors.

9 References

Anson, M. (2010). The handbook of traditional and alternative investment vehicles: Investment characteristics and strategies. Wiley.

Antaki, Marc. (February 4, 2025) "Investment Companies Can Use AI Responsibly to Gain an Edge." How AI Can Redefine Investment Strategy and Generate Value for Financial Firms, World Economic Forum. Retrieved from www.weforum.org/stories/2025/02/ai-redefine-investment-strategy-generate-value-financial-firms/.

Makortoff, Kalyeena. Bank of England says AI software could create market crisis for profit. The Guardian. https://www.theguardian.com/business/2025/apr/09/bank-of-england-says-ai-software-could-create-market-crisis-profit

Altshuller, Stan. Novus. (March 8, 2017). Why hedge funds and investors need data more than ever. Novus. Retrieved from https://www.novus.com/blog/hedge-funds-investors-need-data

CFA Institute. (n.d.). Portfolio construction and investment strategies. CFA Institute Research and Policy Center. Retrieved from https://rpc.cfainstitute.org/topics/portfolio-construction-and-investment

Chadwick. Alex. Novus. (February 22, 2022). The truth about hedge fund marketing. Novus. Retrieved from https://www.novus.com/blog/the-truth-about-hedge-fund-marketing

Emerald Publishing. (2023). A comprehensive literature review on stock market investment from 1900 to 2022. Emerald Insight. Retrieved from https://www.emerald.com/insight/conte

10-2023-0124/full/html

John W. Goodell, Satish Kumar, Weng Marc Lim, Debidutta Pattnaik, Artificial intelligence and machine learning in finance: Identifying foundations, themes, and research clusters from bibliometric analysis, Journal of Behavioral and Experimental Finance, Volume 32, 2021, 100577, ISSN 2214-6350, https://doi.org/10.1016/j.jbef.2021.100577.

Investopedia. (n.d.). Steps to building a profitable investment portfolio. Investopedia. Retrieved from
https://www.investopedia.com/financial-advisor/steps-building-profitable-portfolio

J.P. Morgan Asset Management. (2021). Principles of investing and portfolio diversification. J.P. Morgan. Retrieved from https://am.jpmorgan.com/content/dam/jpm-am-aem/americas/us/en/insights/portfolio-insights/PI-PRINCIPLES_2021.pdf

Litterman, R. (2022). Investment strategies: A practical approach to enhancing investor returns. Springer.

Sivaraman, M. (2023, February 12). Why the 60/40 portfolio may not work anymore. Barron's. Retrieved from https://www.barrons.com/articles/stock-bonds-60-40-portfoli-286a807c

Smith, A. (2024, March 25). Private equity faces challenges amid market shifts. Financial Times. Retrieved from https://www.ft.com/content/23eee6a2-f3c5-474e-a42b-f1cc86134f91

Vanguard. (n.d.). Investor questionnaire. Vanguard. Retrieved from

https://personal.vanguard.com/us/FundsInvQuestionnaire

https://ethz.ch/content/dam/ethz/special-interest/mavt/dynamic-systems-n-control/idsc-dam/L-ectures/Stochastic-Systems/SDE.pdf

10 Code Appendix

10.1 Code for Modeling Single Stock

```
import numpy as np
import math


def simulate_stock_price_with_drops(S0, mu, sigma, T, dt, drop_prob, min_drop, max_dr
  """
  Simulates a stock price path using a log-normal model with random drops.


  Args:
    S0: Initial stock price.
    mu: Annual drift (expected return).
    sigma: Annual volatility.
    T: Time horizon in years.
    dt: Time step (e.g., 1/365 for daily).
    drop_prob: Probability of a drop occurring at each time step.
    min_drop: Minimum percentage drop (e.g., 0.10 for 10%).
    max_drop: Maximum percentage drop (e.g., 0.25 for 25%).


  Returns:
    A tuple containing:
      - time: NumPy array of time points.
      - prices: NumPy array of simulated stock prices.
  """
```

```
N = int(T / dt)  # Number of time steps
time = np.linspace(0, T, N + 1)
prices = np.zeros(N + 1)
prices[0] = S0

# Pre-calculate constants
drift_term = (mu - 0.5 * sigma**2) * dt
vol_term = sigma * math.sqrt(dt)

for t in range(1, N + 1):
  # Generate standard normal random variable
  Z = np.random.standard_normal()

  # Calculate log-normal price change (without drop)
  log_return = drift_term + vol_term * Z
  price_temp = prices[t-1] * math.exp(log_return)

  # Check for random drop event
  if np.random.uniform(0,1) < drop_prob:
    # Generate random drop percentage
    drop_percentage = np.random.uniform(min_drop, max_drop)
    # Apply drop
    prices[t] = price_temp * (1 - drop_percentage)
    # print(f"Time {time[t]:.4f}: Drop occurred! Percentage: {drop_percentage:.2%},
    Price: {prices[t]:.2f}") # Optional: print drop events
  else:
```

```
        # No drop
        prices[t] = price_temp

    return time, prices

# Inputting Specific Parameters
initial_price = 100     # desired S0
annual_drift = 0.10     # desired expected yearly return
annual_volatility = 0.20 # desired standard dev
time_horizon = 40        # No. of years to model
daily_time_step = 1/365 # dt
drop_probability = 1/5000 # As specified
min_drop_percent = 0.10
max_drop_percent = 0.25

# Run the simulation
time_points, stock_prices = simulate_stock_price_with_drops(
    S0=initial_price,
    mu=annual_drift,
    sigma=annual_volatility,
    T=time_horizon,
    dt=daily_time_step,
    drop_prob=drop_probability,
    min_drop=min_drop_percent,
    max_drop=max_drop_percent
)
```

```
# Plotting the results
try:
    import matplotlib.pyplot as plt
    plt.figure(figsize=(10, 6))
    plt.plot(time_points, stock_prices)
    plt.title(f'Simulated Stock Price Path (S0={initial_price}, mu={annual_drift},
    sigma={annual_volatility}) with Random Drops')
    plt.xlabel('Time (Years)')
    plt.ylabel('Stock Price')
    plt.grid(True)
    plt.show()
except ImportError:
    print("Matplotlib not installed. Cannot plot the results.")
    print("\nSample final price:", stock_prices[-1])

# Print first few and last few prices as an example
print("\n--- Sample Simulated Prices ---")
print("Time (Years) | Price")
print("-------------------------")
for i in list(range(5)) + list(range(len(stock_prices)-5, len(stock_prices))):
  print(f"{time_points[i]:<12.4f} | {stock_prices[i]:.2f}")
```

10.2 Code for confidence intervals

```
    import numpy as np
import math
import time as timer # Rename to avoid conflict with time array
import matplotlib.pyplot as plt # Import directly as we know we need it

# Keep your original simulation function (no changes needed here)
def simulate_stock_price_with_drops(S0, mu, sigma, T, dt, drop_prob, min_drop, max_dr
  """
  Simulates a stock price path using a log-normal model with random drops.

  Args:
    S0: Initial stock price.
    mu: Annual drift (expected return).
    sigma: Annual volatility.
    T: Time horizon in years.
    dt: Time step (e.g., 1/365 for daily).
    drop_prob: Probability of a drop occurring at each time step.
    min_drop: Minimum percentage drop (e.g., 0.10 for 10%).
    max_drop: Maximum percentage drop (e.g., 0.25 for 25%).

  Returns:
    A tuple containing:
      - time: NumPy array of time points.
      - prices: NumPy array of simulated stock prices.
```

```
"""
N = int(T / dt)  # Number of time steps
time_points = np.linspace(0, T, N + 1) # Renamed from 'time'
prices = np.zeros(N + 1)
prices[0] = S0

# Pre-calculate constants
drift_term = (mu - 0.5 * sigma**2) * dt
vol_term = sigma * math.sqrt(dt)

for t in range(1, N + 1):
  # Generate standard normal random variable
  Z = np.random.standard_normal()

  # Calculate log-normal price change (without drop)
  log_return = drift_term + vol_term * Z
  price_temp = prices[t-1] * math.exp(log_return)

  # Check for random drop event
  if np.random.uniform(0,1) < drop_prob:
    # Generate random drop percentage
    drop_percentage = np.random.uniform(min_drop, max_drop)
    # Apply drop
    prices[t] = price_temp * (1 - drop_percentage)
  else:
    # No drop
```

```
      prices[t] = price_temp

  return time_points, prices

# --- Simulation Parameters ---
initial_price = 100     # S0
annual_drift = 0.10     # mu
annual_volatility = 0.20 # sigma
time_horizon = 40        # T
daily_time_step = 1/365 # dt
drop_probability = 1/5000
min_drop_percent = 0.10
max_drop_percent = 0.25

# --- Monte Carlo Parameters ---
num_simulations = 1000   # Number of simulation paths to run
# Define multiple confidence levels and their colors
confidence_intervals = {
    0.95: 'lightblue',  # Widest interval, lightest color
    0.80: 'skyblue',
    0.60: 'deepskyblue' # Narrowest interval, darkest color
}

# --- Run Multiple Simulations ---
print(f"Running {num_simulations} simulations...")
start_time = timer.time()
```

```
# Store all simulation results
num_steps = int(time_horizon / daily_time_step) + 1
all_prices = np.zeros((num_steps, num_simulations))
time_points = None # To store the time axis once

for i in range(num_simulations):
    # Run one simulation
    sim_time, sim_prices = simulate_stock_price_with_drops(
        S0=initial_price,
        mu=annual_drift,
        sigma=annual_volatility,
        T=time_horizon,
        dt=daily_time_step,
        drop_prob=drop_probability,
        min_drop=min_drop_percent,
        max_drop=max_drop_percent
    )
    # Store the results
    all_prices[:, i] = sim_prices
    if time_points is None:
        time_points = sim_time # Store time axis from the first run

    # Optional: Print progress
    if (i + 1) % (num_simulations // 10) == 0:
      print(f"  Completed {i + 1}/{num_simulations} simulations...")
```

```
end_time = timer.time()
print(f"Simulations complete. Time taken: {end_time - start_time:.2f} seconds.")

# --- Calculate Median and Confidence Intervals ---
print("Calculating percentiles...")
median_prices = np.percentile(all_prices, 50, axis=1)
interval_bounds = {} # Dictionary to store bounds for each CI level

# Sort levels to plot widest first
sorted_levels = sorted(confidence_intervals.keys(), reverse=True)

for level in sorted_levels:
    lower_percentile = (1 - level) / 2 * 100
    upper_percentile = (1 + level) / 2 * 100
    lower_bound = np.percentile(all_prices, lower_percentile, axis=1)
    upper_bound = np.percentile(all_prices, upper_percentile, axis=1)
    interval_bounds[level] = (lower_bound, upper_bound)
    print(f"  Calculated {int(level*100)}% CI (Percentiles: {lower_percentile:.1f}%,


# --- Plotting the Results ---
try:
    plt.figure(figsize=(12, 7))
```

```
    # Plot the shaded confidence intervals (widest first)
    for level in sorted_levels:
        lower_bound, upper_bound = interval_bounds[level]
        color = confidence_intervals[level]
        plt.fill_between(time_points, lower_bound, upper_bound, color=color, alpha=0.
                         label=f'{int(level*100)}% Confidence Interval')

    # Plot the median path on top
    plt.plot(time_points, median_prices, label='Median Path (50th Percentile)', color

    plt.title(f'Simulated Stock Price Paths ({num_simulations} Runs) with Confidence
              f'S0={initial_price}, mu={annual_drift}, sigma={annual_volatility}, T={
    plt.xlabel('Time (Years)')
    plt.ylabel('Stock Price')
    plt.yscale('log') # Log scale often useful for long-term stock prices
    plt.grid(True, which="both", ls="--", linewidth=0.5)
    plt.legend()
    plt.show()

except ImportError:
    # This part is less likely to be reached now since we import plt at the top,
    # but kept for robustness if the import fails unexpectedly later.
    print("Matplotlib not installed or failed to import. Cannot plot the results.")
    print("\nSample final price statistics:")
    print(f"Median final price: {median_prices[-1]:.2f}")
    for level in sorted_levels:
```

```
        lower, upper = interval_bounds[level]
        print(f"{int(level*100)}% CI for final price: ({lower[-1]:.2f}, {upper[-1]:.

# Print first few and last few values for the bounds and median
print("\n--- Sample Median and 95% CI Bounds ---") # Just show 95% bounds for brevity
print("Time (Years) | 2.5% Bound | Median Price | 97.5% Bound")
print("-------------------------------------------------------")
lower_95, upper_95 = interval_bounds[0.95]
indices_to_show = list(range(5)) + list(range(len(time_points)-5, len(time_points)))
for i in indices_to_show:
  print(f"{time_points[i]:<12.4f} | {lower_95[i]:<10.2f} | {median_prices[i]:<12.2f}
```

10.3 60/40 portfolio code

```
    import numpy as np
import math
import time as timer
import matplotlib.pyplot as plt

# --- Asset Simulation Logic ---

def simulate_portfolio_path(S0_portfolio, asset_params, weights, T, dt):
    """
    Simulates the value path of a portfolio composed of multiple assets.
```

```
Args:
    S0_portfolio: Initial total portfolio value.
    asset_params: A list of dictionaries, each containing params for an asset:
                  {'mu': annual_drift, 'sigma': annual_volatility,
                   'drop_prob': probability_of_drop (0 for bonds),
                   'min_drop': min_drop_percentage (0 for bonds),
                   'max_drop': max_drop_percentage (0 for bonds)}
    weights: A list of portfolio weights for each asset (must sum to 1).
    T: Time horizon in years.
    dt: Time step (e.g., 1/365 for daily).

Returns:
    A tuple containing:
      - time_points: NumPy array of time points.
      - portfolio_values: NumPy array of simulated total portfolio values.
"""
num_assets = len(asset_params)
if len(weights) != num_assets:
    raise ValueError("Number of weights must match number of asset parameter sets
if not np.isclose(sum(weights), 1.0):
     raise ValueError("Weights must sum to 1.0")

N = int(T / dt)  # Number of time steps
time_points = np.linspace(0, T, N + 1)

# Initialize price arrays for each asset
```

```
asset_prices = np.zeros((num_assets, N + 1))
for i in range(num_assets):
    asset_prices[i, 0] = S0_portfolio * weights[i] # Initial value for each asset

# Pre-calculate constants for each asset
drift_terms = np.zeros(num_assets)
vol_terms = np.zeros(num_assets)
drop_probs = np.zeros(num_assets)
min_drops = np.zeros(num_assets)
max_drops = np.zeros(num_assets)

for i in range(num_assets):
    params = asset_params[i]
    mu = params['mu']
    sigma = params['sigma']
    drift_terms[i] = (mu - 0.5 * sigma**2) * dt
    vol_terms[i] = sigma * math.sqrt(dt)
    drop_probs[i] = params.get('drop_prob', 0) # Default to 0 if not provided
    min_drops[i] = params.get('min_drop', 0)
    max_drops[i] = params.get('max_drop', 0)

# --- Simulation Loop ---
for t in range(1, N + 1):
    for i in range(num_assets):
        # Get previous price
        prev_price = asset_prices[i, t-1]
```

```
# Skip calculation if previous price is zero or negative (can happen with
if prev_price <= 0:
    asset_prices[i, t] = 0
    continue

# Generate standard normal random variable (only if needed)
Z = 0
if vol_terms[i] > 0 or drop_probs[i] > 0: # Only generate randomness if s
     Z = np.random.standard_normal()

# Calculate log-normal price change (if applicable)
if vol_terms[i] > 0: # Stochastic component exists
    log_return = drift_terms[i] + vol_terms[i] * Z
    price_temp = prev_price * math.exp(log_return)
else: # Deterministic component (like bonds)
    price_temp = prev_price * math.exp(drift_terms[i]) # Simplified for s

# Check for random drop event (only for assets with drop_prob > 0)
if drop_probs[i] > 0 and np.random.uniform(0, 1) < drop_probs[i]:
    # Generate random drop percentage
    drop_percentage = np.random.uniform(min_drops[i], max_drops[i])
    # Apply drop
    asset_prices[i, t] = price_temp * (1 - drop_percentage)
else:
    # No drop or not applicable
```

```
                asset_prices[i, t] = price_temp

            # Ensure price doesn't go below zero
            if asset_prices[i, t] < 0:
                 asset_prices[i, t] = 0



    # Calculate total portfolio value at each time step
    portfolio_values = np.sum(asset_prices, axis=0)

    return time_points, portfolio_values

# --- Portfolio & Simulation Parameters ---
initial_portfolio_value = 100 # S0 for the whole portfolio

# Define asset parameters as a list of dictionaries
asset_parameters = [
    {'mu': 0.15, 'sigma': 0.40, 'drop_prob': 1/5000, 'min_drop': 0.10, 'max_drop': 0.
    {'mu': 0.11, 'sigma': 0.20, 'drop_prob': 1/5000, 'min_drop': 0.10, 'max_drop': 0.
    {'mu': 0.07, 'sigma': 0.10, 'drop_prob': 1/5000, 'min_drop': 0.10, 'max_drop': 0.
    {'mu': 0.04, 'sigma': 0.00, 'drop_prob': 0, 'min_drop': 0, 'max_drop': 0}
]
portfolio_weights = [0.20, 0.20, 0.20, 0.40] # Weights corresponding to the assets ab

time_horizon = 40        # T
daily_time_step = 1/365 # dt
```

```
# --- Monte Carlo Parameters ---
num_simulations = 1000   # Number of simulation paths to run
# Define multiple confidence levels and their colors
confidence_intervals = {
    # Level: (Color for Fill, Percentile Label Color) - Using slightly different colo
    0.95: ('paleturquoise', 'darkcyan'),   # Widest interval, lightest color
    0.80: ('lightskyblue', 'mediumblue'),
    0.60: ('cornflowerblue', 'darkblue')  # Narrowest interval, darker color
}

# --- Run Multiple Simulations ---
print(f"Running {num_simulations} portfolio simulations...")
start_time = timer.time()

# Store all portfolio simulation results
num_steps = int(time_horizon / daily_time_step) + 1
all_portfolio_values = np.zeros((num_steps, num_simulations))
time_points = None # To store the time axis once

for i in range(num_simulations):
    # Run one portfolio simulation
    sim_time, sim_portfolio_values = simulate_portfolio_path(
        S0_portfolio=initial_portfolio_value,
        asset_params=asset_parameters,
        weights=portfolio_weights,
```

```
        T=time_horizon,
        dt=daily_time_step
    )
    # Store the results
    all_portfolio_values[:, i] = sim_portfolio_values
    if time_points is None:
        time_points = sim_time # Store time axis from the first run

    # Optional: Print progress
    if (i + 1) % (num_simulations // 10) == 0:
      print(f"  Completed {i + 1}/{num_simulations} simulations...")


end_time = timer.time()
print(f"Simulations complete. Time taken: {end_time - start_time:.2f} seconds.")

# --- Calculate Median and Confidence Intervals for the Portfolio ---
print("Calculating portfolio percentiles...")
median_portfolio_values = np.percentile(all_portfolio_values, 50, axis=1)
portfolio_interval_bounds = {} # Dictionary to store bounds for each CI level

# Sort levels to plot widest first
sorted_levels = sorted(confidence_intervals.keys(), reverse=True)

for level in sorted_levels:
    lower_percentile = (1 - level) / 2 * 100
```

```
    upper_percentile = (1 + level) / 2 * 100
    lower_bound = np.percentile(all_portfolio_values, lower_percentile, axis=1)
    upper_bound = np.percentile(all_portfolio_values, upper_percentile, axis=1)
    portfolio_interval_bounds[level] = (lower_bound, upper_bound)
    print(f"  Calculated {int(level*100)}% CI (Percentiles: {lower_percentile:.1f}%,


# --- Plotting the Results ---
try:
    plt.figure(figsize=(12, 7))

    # Plot the shaded confidence intervals (widest first)
    for level in sorted_levels:
        lower_bound, upper_bound = portfolio_interval_bounds[level]
        fill_color, _ = confidence_intervals[level] # Get fill color
        plt.fill_between(time_points, lower_bound, upper_bound, color=fill_color, alp
                         label=f'{int(level*100)}% Confidence Interval')

    # Plot the median path on top
    plt.plot(time_points, median_portfolio_values, label='Median Portfolio Path (50th

    # Construct title with portfolio details
    asset_desc = [f"{w*100:.0f}% (={p['mu']:.2f}, ={p['sigma']:.2f})" for w, p in zip
    title_str = f'Simulated Portfolio Value Paths ({num_simulations} Runs) with Confi
    title_str += f'Initial Value: ${initial_portfolio_value}, T={time_horizon} yrs\n'
    title_str += f'Assets: [{"; ".join(asset_desc)}]'
```

```
    plt.title(title_str, fontsize=10) # Adjust fontsize if needed
    plt.xlabel('Time (Years)')
    plt.ylabel('Portfolio Value')
    plt.yscale('log') # Log scale is highly recommended for portfolio values over lon
    plt.grid(True, which="both", ls="--", linewidth=0.5)
    plt.legend()
    plt.tight_layout() # Adjust layout to prevent labels overlapping
    plt.show()

except ImportError:
    print("Matplotlib not installed or failed to import. Cannot plot the results.")
    print("\nSample final portfolio value statistics:")
    print(f"Median final value: {median_portfolio_values[-1]:.2f}")
    for level in sorted_levels:
         lower, upper = portfolio_interval_bounds[level]
         print(f"{int(level*100)}% CI for final value: ({lower[-1]:.2f}, {upper[-1]:.

# Print first few and last few values for the bounds and median
print("\n--- Sample Median Portfolio Value and 95% CI Bounds ---") # Just show 95% bo
print("Time (Years) | 2.5% Bound | Median Value | 97.5% Bound")
print("-------------------------------------------------------")
lower_95, upper_95 = portfolio_interval_bounds[0.95]
indices_to_show = list(range(5)) + list(range(len(time_points)-5, len(time_points)))
for i in indices_to_show:
```

```
  print(f"{time_points[i]:<12.4f} | {lower_95[i]:<10.2f} | {median_portfolio_values[i
```

10.4 Efficient Frontier Portfolio Design

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns
from scipy.stats import norm
import yfinance as yf
import cvxpy as cp

etf_tickers_NASDAQ = ['^IXIC','VGT', 'XLF', 'XLY', 'XLV', 'VOX', 'XLI', 'XLP', 'XLE',
etf_tickers = ['VGT', 'XLF', 'XLY', 'XLV', 'VOX', 'XLI', 'XLP', 'XLE', 'VNQ', 'XLB',
mf_tickers_NASDAQ = ['^IXIC', 'VITAX', 'VFAIX', 'BPTRX', 'VGHCX', 'VTCAX', 'VINAX', '
mf_tickers = ['VITAX', 'VFAIX', 'BPTRX', 'VGHCX', 'VTCAX', 'VINAX', 'VCSAX', 'VENAX',

start_date_15yr = '2010-01-01'
end_date = '2025-01-04'

day = '1d'
week = '1wk'
month = '1mo'
year = '1yr'

def import_stock_data(tickers, start_date, end_date, frequency):
```

```
  data = yf.download(tickers, start=start_date, end = end_date, interval=frequency, a
  data=data['Close']
  data = data.dropna()
  data = data.reindex(columns=tickers)
  names = tickers
  data.index = pd.to_datetime(data.index)
  return(data)

def annualize_data(data):
    data.index = pd.to_datetime(data.index)
    annual_data = data.groupby(data.index.year).first()
    annual_data.index = pd.to_datetime(annual_data.index.astype(str) + '-01-01')
    return annual_data

def log_returns(data):
    return (np.log(1+data.pct_change()))

def prices_cov_matrix(data):
  return(data.cov())

def returns_cov_matrix(data):
  data=log_returns(data)
  return(data.cov())

def summary_table(data, frequency):
  data=log_returns(data).dropna()
```

```
total_return = np.exp(data.sum()) - 1
num_years = (data.index[-1] - data.index[0]) / pd.Timedelta(days=365.25)
annual_return = (total_return+1)**(1/num_years)-1
variance = data.var()
volatility = variance**(1/2)

if frequency == day:
  scale=252
elif frequency ==week:
  scale=52.18
elif frequency ==month:
  scale=12
elif frequency == year:
  scale=1
annualized_variance = variance*scale
annualized_volatility = annualized_variance**(1/2)

summary = pd.DataFrame({
    'Ticker': data.columns,
    'Total Return': (total_return.values)*100,
    'Yearly Return': (annual_return.values)*100,
    'Volatilty': volatility*100,
    'Annualized Volatility': annualized_volatility*100
})
return(summary)
```

```
def optimization_matrix(data, frequency):
    returns_vector = summary_table(data, frequency)['Yearly Return'].values / 100
    returns_vector = returns_vector.reshape(-1, 1)
    c = returns_cov_matrix(data).values
    n=c.shape[0]
    top=np.hstack([c, -returns_vector, -np.ones((n,1))])
    middle = np.hstack([returns_vector.T, np.array([[0]]), np.array([[0]])])
    bottom = np.hstack([np.ones((1,n)), np.array([[0]]), np.array([[0]])])
    final_matrix = np.vstack([top, middle, bottom])
    final_matrix=pd.DataFrame(final_matrix)
    return(final_matrix)

def optimization(data, frequency, mu):
    a = optimization_matrix(data, frequency)
    n=a.shape[0]
    b = np.zeros((n,1))
    b[-2]=mu
    b[-1]=1
    x = np.linalg.solve(a,b)
    weights=x[:-3]
    weights = pd.DataFrame(weights)
    return(weights)

def nonnegative_optimization(data, frequency, mu):
    summary = summary_table(data, frequency)
    r = summary['Yearly Return'].values / 100  # expected returns
```

```
Sigma = returns_cov_matrix(data).values            # covariance matrix
n = len(r)
x = cp.Variable(n)
constraints = [
    r @ x == mu,     # target return
    cp.sum(x) == 1,  # fully invested
    x >= 0           # long-only (no shorting)
]
objective = cp.Minimize(cp.quad_form(x, Sigma))
problem = cp.Problem(objective, constraints)
problem.solve()
if problem.status != cp.OPTIMAL:
    raise ValueError("Optimization did not converge")
weights=x.value
if frequency == day:
  scale=252
elif frequency ==week:
  scale=52.18
elif frequency ==month:
  scale=12
elif frequency == year:
  scale=1
variance = weights.T @ Sigma @ weights
volatility = variance**(1/2)
annualized_variance = variance*scale
annualized_volatility = annualized_variance**(1/2)
```

```
    weights=weights*100
    weights = pd.DataFrame(weights, index=summary['Ticker'], columns=["Weight"])
    volatilatiies = pd.DataFrame({
        "Weight": [volatility.item()*100, annualized_volatility.item()*100]
    }, index=["Volatility", "Annualized Volatility"])
    weights = pd.concat([weights, volatilatiies])
    return weights

def multi_optimization(data, frequency, mu_list):
  results=[]
  for mu in mu_list:
      portfolio = nonnegative_optimization(data, frequency, mu)
      portfolio = portfolio.rename(columns={"Weight": mu*100})
      results.append(portfolio)
  combined = pd.concat(results, axis=1)
  combined.index.name = "Ticker"
  combined.reset_index(inplace=True)
  return combined

def efficient_frontier(data, frequency):
  summary = summary_table(data, frequency)
  returns = summary['Yearly Return'] / 100
  mu_min = returns.min()
  mu_max = returns.max()
  mu_values = np.linspace(mu_min, mu_max, 200)
  volatilities = []
```

```
  yearly_returns = []

  for mu in mu_values:
        portfolio = nonnegative_optimization(data, frequency, mu)
        annualized_volatility = portfolio.loc["Annualized Volatility", "Weight"]  # c
        yearly_return = mu * 100  # mu is in decimal, need to match axis in %

        volatilities.append(annualized_volatility)
        yearly_returns.append(yearly_return)
  plt.plot(volatilities, yearly_returns, marker = 'o', linestyle='-', color='b', mark
  plt.title('Efficient Frontier', fontsize=14)
  plt.ylabel('Yearly Return (%)', fontsize=12)
  plt.xlabel('Annualized Volatility (%)', fontsize=12)
  plt.show()

def efficient_frontier_cml(data, frequency, risk_free_rate):
  summary = summary_table(data, frequency)
  returns = summary['Yearly Return'] / 100
  mu_min = returns.min()
  mu_max = returns.max()
  mu_values = np.linspace(mu_min, mu_max, 200)
  volatilities = []
  yearly_returns = []

  for mu in mu_values:
        portfolio = nonnegative_optimization(data, frequency, mu)
```

```
    annualized_volatility = portfolio.loc["Annualized Volatility", "Weight"]  # c
    yearly_return = mu * 100  # mu is in decimal, need to match axis in %
    volatilities.append(annualized_volatility)
    yearly_returns.append(yearly_return)
volatilities = np.array(volatilities)
yearly_returns = np.array(yearly_returns)
sharpe_ratios = (yearly_returns/100 - risk_free_rate) / (volatilities/100)
min_volatility_idx = volatilities.argmin()
min_volatility = volatilities[min_volatility_idx]
min_volatility_return = yearly_returns[min_volatility_idx]
max_sharpe_idx = sharpe_ratios.argmax()
max_sharpe_volatility = volatilities[max_sharpe_idx]
max_sharpe_return = yearly_returns[max_sharpe_idx]
cml_x = np.linspace(0, max_sharpe_volatility+5, 100)
cml_y = risk_free_rate*100 + ( (max_sharpe_return - risk_free_rate*100) / max_sharp

plt.figure()
plt.plot(volatilities, yearly_returns, marker = 'o', linestyle='-', color='b', mark
plt.plot(cml_x, cml_y, color='r', linestyle='--', label='Capital Market Line')
plt.scatter(max_sharpe_volatility, max_sharpe_return, color='g', s=100, marker='*',
plt.scatter(min_volatility, min_volatility_return, color='m', s=100, marker='*', la

plt.title(f'Efficient Frontier with Capital Market Line with Risk-Free Rate {risk_f
plt.xlabel('Annualized Volatility (%)', fontsize=12)
plt.ylabel('Yearly Return (%)', fontsize=12)
plt.legend()
```

```
  plt.grid(True)
  plt.show()
  optimal_portfolio = multi_optimization(data, frequency, (max_sharpe_return/100, min
  return optimal_portfolio


etf_tickers_data_1mo_15yr = import_stock_data(etf_tickers, start_date_15yr, end_date,

#print(summary_table(etf_tickers_data_1mo_15yr, month))

#print(multi_optimization(etf_tickers_data_1mo_15yr, month, (0.07, 0.08, 0.09, 0.1, 0

print(efficient_frontier_cml(etf_tickers_data_1mo_15yr, month, 0.0432))

mf_tickers_data_1mo_15yr = import_stock_data(mf_tickers, start_date_15yr, end_date, m
mf_tickers_NASDAQ_data_1mo_15yr = import_stock_data(mf_tickers_NASDAQ, start_date_15y

#print(summary_table(mf_tickers_NASDAQ_data_1mo_15yr, month))
#print(multi_optimization(mf_tickers_data_1mo_15yr, month, (0.075, 0.1, 0.125, 0.15))
#print(efficient_frontier(mf_tickers_data_1mo_15yr, month))
print(efficient_frontier_cml(mf_tickers_data_1mo_15yr, month, 0.0432))
```

11 AI Appendix

We asked a few AIs two questions about investing in the long run and recorded here their responses. The AIs queried were ChatGPT, Grok, Gemini, Perplexity, and Copilot.

11.1 Question 1: What average yearly return and variance should I strive for in my long-run investing portfolio over a 20 year period?

ChatGPT

Average Yearly Return (Target): Moderate to Aggressive Portfolio (60-80% equities):

Historical average return: 6% to 8% annually after inflation

Nominal returns (before inflation): 8% to 10%

This is consistent with historical SP 500 returns, which have averaged about 10–11% nominally over long periods. Adding bonds and diversifying lowers risk and slightly lowers expected return.

Variance / Standard Deviation (Risk): Moderate risk portfolio (e.g., 60/40 stocks/bonds):

Standard deviation: 10% to 12%

Aggressive portfolio (e.g., 80–100% stocks):

Standard deviation: 15% to 20%

Variance is the square of standard deviation. So if your SD is 15%, the variance is $0.15^2 = 0.0225$ (or 2.25%).

Practical Long-Run Targets (for a balanced investor): Target annual return: 7% nominal (5% real return)

Target variance: 0.01–0.02

Sharpe Ratio (return per unit of risk): 0.5 is solid; 0.7–1.0+ is excellent

Grok

Final Recommendation For a 20-year horizon with a balanced approach:

Target Average Yearly Return: Aim for 6-8% nominal (4-6% real) annually. Acceptable Variance: Expect annual returns to fluctuate between -10% and +15%, with occasional worse years during market downturns. Over 20 years, the compounding effect should smooth out volatility, provided you stay invested. If you want a more tailored answer, provide details about your current portfolio, risk tolerance, and specific goals, and I can help refine this further. Alternatively, I can search for the latest market data or trends as of April 2025 to adjust these estimates. Let me know!

Gemini

What Should You Strive For?

Instead of a fixed number, consider a range based on your risk profile:

Aggressive Investor (High Risk Tolerance): Might aim for returns closer to the historical stock market average (e.g., 8-10%+ nominal annually), accepting the associated higher variance (e.g., 15%+ standard deviation). This usually involves a portfolio heavily weighted towards stocks. Moderate Investor (Medium Risk Tolerance): Might aim for returns between the stock market and a balanced portfolio (e.g., 7-9% nominal annually), with moderate variance (e.g., 10-14% standard deviation). This often aligns with portfolios like the 60/40 split. Conservative Investor (Low Risk Tolerance): Might prioritize capital preservation, aiming for lower returns (e.g., 4-6% nominal annually) with lower variance (e.g., below 10% standard deviation). This typically involves a higher allocation to bonds and cash equivalents.

Recommendation:

Assess Your Risk Tolerance: Be honest about how much volatility you can comfortably handle, both financially and emotionally. Consider using online questionnaires or consulting a financial advisor. Define Your Financial Goals: What is this money for? Retirement, a house down payment (though 20 years is long for this), etc.? The importance of the goal can influence risk tolerance. Use Historical Data as a Guide: Understand the typical risk/return profiles of different asset allocations (like 100% stocks vs. 60/40). Set Realistic Expectations: Remember that past performance doesn't guarantee future results, and achieving higher returns always involves taking on more risk. By considering these factors, you can establish return and variance goals that align with your personal situation and increase your chances of staying invested for the full 20 years.

Perplexity

Practical Recommendations Base allocation on risk tolerance:

Aggressive: 80-100% stocks (targeting 7.6-10.4% returns).

Moderate: 60% stocks/40% bonds (targeting 8.8% returns).

Reinvest dividends: Accounts for 40% of total SP 500 returns over 20 years.

Rebalance annually: Maintains target allocation and controls variance.

For most investors, a 7-9% inflation-adjusted return with annual volatility under 15% (achievable via diversification) strikes a prudent balance between growth and risk over two decades

Copilot

If you're comfortable with a higher level of risk, you might aim for average annual returns in the 7%–10% range, acknowledging that the corresponding annual volatil-

ity could be around 15%–20% (yielding a variance roughly between 0.0225 and 0.04). For a more balanced approach, you might expect between 6%–8% return with lower volatility (around 10%–15% standard deviation, and variance roughly between 0.01 and 0.0225). These numbers are starting points—the right target for you is ultimately based on your personal financial situation and risk appetite.

11.2 Question 2: What types of assets should I include in my long-run investing portfolio?

ChatGPT

Core Asset Classes 1. Equities (Stocks) – Growth Engine US Stocks (SP 500, Russell 2000)

Strong historical returns (10% long-term)

Consider total market or SP 500 index funds

International/Developed Markets

Diversifies geography and currency risk

Look at MSCI EAFE or similar

Emerging Markets

Higher risk, higher return potential

Great for long-term growth

Target allocation: 60–80% for long-term investors Use low-cost index ETFs or mutual funds

2. Fixed Income (Bonds) – Stability Income US Treasuries, Corporate Bonds

Lower return, but buffer in down markets

Inflation-Protected Bonds (TIPS)

Protect purchasing power

International Bonds (optionally, currency-hedged)

Target allocation: 10–30% Choose duration based on your risk tolerance (shorter = safer)

3. Real Assets – Inflation Hedge Real Estate (REITs or physical)

Long-term growth + income

Low correlation with stocks

Commodities (optional)

Oil, gold, agriculture — more volatile, but hedge inflation

Target allocation: 5–15% REIT ETFs are easy ways to add real estate exposure

4. Alternative Investments – Optional, High Risk/Reward Private equity, venture capital, hedge funds

Often require accredited investor status

Less liquid, higher fees

Crypto (Bitcoin, Ethereum, etc.)

Highly speculative, but long-term upside

Only if you're comfortable with big swings

Target allocation: 0–5% (if at all) Only with money you can afford to lose

Tips for Long-Term Portfolio Construction Diversify globally – Don't put all your faith in one country or market

Use low-fee index funds or ETFs – Maximize compounding

Rebalance annually – Keep your risk in check

Dollar-cost average – Smooth out the impact of volatility

Stay invested – Time in the market beats timing the market

Would you like me to sketch out a sample portfolio (e.g., conservative, balanced, or aggressive)? Or tailor one based on your age, risk tolerance, or retirement target?

Grok

As of April 14, 2025, building a long-run investing portfolio requires a balanced approach that considers diversification, risk tolerance, and financial goals. Here are the key types of assets you should consider including, along with some insights based on current trends and best practices:

1. Equities (Stocks) Why include? Stocks offer long-term growth potential and are a core component of most portfolios. They can provide capital appreciation and dividends. Types to consider: Large-cap stocks: Stable, well-established companies (e.g., Apple, Microsoft) for reliability. Mid-cap and small-cap stocks: Higher growth potential but with more risk. International stocks: Exposure to global markets, especially emerging markets like India or Southeast Asia, which are showing strong growth as of 2025. Exchange-Traded Funds (ETFs) and Mutual Funds: For diversification, consider broad-market ETFs (e.g., SP 500) or sector-specific funds (e.g., technology, healthcare). Current trend: In 2025, tech and renewable energy sectors remain strong, but inflation and interest rate changes could impact valuations. Diversify across sectors to mitigate risk.

2. Fixed Income (Bonds) Why include? Bonds provide stability and income, acting as a buffer against stock market volatility. They're essential for reducing overall portfolio risk. Types to consider: Government bonds: U.S. Treasuries or municipal bonds for safety. Corporate bonds: Higher yields but with more risk. Bond ETFs: For diversification and ease of management. Current trend: Interest rates in 2025 are stabilizing after recent fluctuations. Longer-term bonds may offer higher yields, but watch for inflation risks.

3. Real Estate Why include? Real estate provides inflation protection and passive

income (e.g., rental properties) or appreciation. Types to consider: Physical property: Residential or commercial real estate. Real Estate Investment Trusts (REITs): Publicly traded companies that own and operate real estate, offering liquidity and diversification. Current trend: As of 2025, urban real estate markets are recovering, but regional differences exist. REITs are popular for their dividend yields and lower entry barriers.

4. Cash and Cash Equivalents Why include? Cash provides liquidity for opportunities or emergencies and reduces risk during market downturns. Types to consider: High-yield savings accounts: For safety and interest. Money market funds: Low risk, highly liquid. Current trend: With interest rates stabilizing, cash equivalents are offering competitive returns compared to recent years.

5. Alternative Investments Why include? Alternatives can enhance returns and reduce correlation with traditional assets, though they often carry higher risk. Types to consider: Commodities: Gold, silver, or oil for inflation hedging. Cryptocurrencies: Bitcoin and Ethereum are gaining legitimacy, but they're volatile. In 2025, regulatory clarity is improving. Private equity or venture capital: For higher risk/reward, especially in tech or green energy. Collectibles or Art: Niche but can appreciate over decades. Current trend: Crypto markets are maturing, but only allocate a small percentage (e.g., 5%) due to volatility. Green energy investments are particularly hot in 2025.

6. Retirement Accounts and Tax-Advantaged Assets Why include? Maximize tax benefits and compound growth over time. Types to consider: 401(k) or IRA: Contribute to employer-sponsored plans or individual retirement accounts. Roth accounts: Pay taxes now for tax-free withdrawals later. Current trend: Tax laws in 2025 favor long-term retirement savings, with incentives for sustainable investments. Key Considerations for 2025: Diversification: Don't put all your eggs in one basket.

A mix of asset classes reduces risk. Risk Tolerance: Younger investors can handle more equities, while those nearing retirement should lean toward bonds and cash. Fees and Costs: Look for low-cost index funds or ETFs to minimize expenses. Sustainability: ESG (Environmental, Social, Governance) investing is gaining traction. Companies with strong sustainability practices may outperform in the long run. Rebalancing: Review and adjust your portfolio annually or after major life events to maintain your desired asset allocation. Inflation and Interest Rates: In 2025, inflation is expected to be moderate, but keep an eye on geopolitical risks and central bank policies. Sample Portfolio Allocation (Long-Term, Moderate Risk): 60% Equities (50% U.S., 10% International) 20% Fixed Income (Bonds) 10% Real Estate (REITs or direct investment) 5% Cash/Cash Equivalents 5% Alternatives (e.g., crypto, commodities) Final Tip: Consult with a financial advisor to tailor this to your specific situation, including your age, income, goals, and tax status. If you have specific companies, funds, or trends in mind (e.g., AI-driven stocks or green bonds), I can help analyze them further or search for additional data. Let me know if you'd like more detailed analysis or have questions about current market conditions!

Gemini

Okay, building on the idea of a long-term (e.g., 20-year) portfolio, choosing the right mix of asset types is crucial for balancing growth potential with risk management. Diversification across different asset classes is key because they often react differently to economic conditions, helping to smooth out returns over time.

Here are the main types of assets typically included in a well-diversified, long-run portfolio:

Equities (Stocks):

Role: Primary engine for growth over the long term. Historically, stocks have provided the highest returns, outpacing inflation significantly, although they come

with higher volatility. Types to Include: Domestic Stocks (U.S.): Include a mix of company sizes (market capitalization) for diversification within the US market: Large-Cap: Stocks of large, established companies (like those in the SP 500). Generally considered less volatile than smaller companies. Mid-Cap: Stocks of medium-sized companies. Offer a blend of growth potential and stability. Small-Cap: Stocks of smaller companies. Higher growth potential but also higher risk and volatility.

International Stocks: Provide geographic diversification, reducing reliance solely on the US economy and capturing growth opportunities abroad. Developed Markets: Stocks from established economies (e.g., Europe, Japan, Canada, Australia). Emerging Markets: Stocks from developing economies (e.g., China, India, Brazil). Higher growth potential, but often higher political and currency risk.

How to Invest: Individual stocks, Stock Mutual Funds, or Exchange-Traded Funds (ETFs). Funds and ETFs offer instant diversification. Fixed Income (Bonds):

Role: Provide stability, generate income (through interest payments), lower overall portfolio volatility, and preserve capital. Bonds often (but not always) perform well when stocks are down. Types to Include: Government Bonds: U.S. Treasuries: Considered very low-risk. Maturities range from short-term (T-bills) to long-term (T-bonds). TIPS (Treasury Inflation-Protected Securities): Principal adjusts with inflation, helping protect purchasing power. Corporate Bonds: Issued by companies. Offer higher yields than government bonds but carry more credit risk. Can range from investment-grade (safer) to high-yield/junk (riskier). Municipal Bonds ("Munis"): Issued by state and local governments. Interest is often exempt from federal income tax (and sometimes state/local tax if you live in the issuing state), making them attractive for taxable accounts. International Bonds: Offer diversification across different countries and interest rate environments. Can be government or corporate. How to Invest: Individual bonds, Bond Mutual Funds, or Bond ETFs.

Funds and ETFs are usually easier for diversification and management. Real Estate:

Role: Potential for appreciation and income (rent), diversification (often has low correlation with stocks/bonds), and can serve as a hedge against inflation. Types to Include: REITs (Real Estate Investment Trusts): Companies that own or finance income-producing real estate. They trade like stocks on major exchanges, providing liquidity and easy diversification across various property types (residential, commercial, industrial, etc.). This is the easiest way for most investors to add real estate exposure. Direct Ownership: Owning physical properties (rental homes, commercial real estate). Requires significant capital, active management, and is relatively illiquid. Commodities (Consider a Smaller Allocation):

Role: Raw materials like oil, natural gas, precious metals (gold, silver), and agricultural products. Can provide diversification (low correlation to stocks/bonds) and act as an inflation hedge. However, they are highly volatile and don't generate income like bonds or dividend stocks. How to Invest: Typically through ETFs or mutual funds that track commodity indices or invest in futures contracts. Investing directly in futures is complex and risky for most individuals. Stocks of commodity-producing companies (e.g., miners, oil companies) offer indirect exposure. Cash and Cash Equivalents:

Role: Primarily for safety, liquidity, and short-term needs (emergency fund). Not a significant part of a long-term growth strategy as inflation erodes its value over time. Types: High-yield savings accounts, money market funds, short-term CDs. Putting it Together (Asset Allocation):

The proportion of your portfolio allocated to each asset class (your asset allocation) is the most critical decision and should be based on your risk tolerance and time horizon (20 years).

A younger investor with a long time horizon and higher risk tolerance might

have a larger allocation to equities (e.g., 70-90%), including significant international exposure, with the remainder in bonds and perhaps a small slice of REITs or commodities. Someone needing more stability or having lower risk tolerance would hold a larger percentage in bonds (e.g., 40-60% or more). For most long-term investors, a core portfolio built around a diversified mix of domestic and international stocks and various types of bonds, potentially supplemented with REITs, forms a solid foundation.

Remember, this is general information. You should consider your specific financial situation and goals, and potentially consult with a qualified financial advisor to determine the best asset allocation for you.

Perplexity

Core Asset Classes Stocks (Equities)

Role: Growth engine of the portfolio due to high potential returns.

Types:

Large-cap stocks: Stable companies with lower risk.

Mid-cap and small-cap stocks: Higher growth potential but increased volatility.

International stocks: Diversification across global markets.

Emerging markets: High-risk, high-reward investments in developing economies.

Bonds (Fixed Income)

Role: Stability and income generation, reducing overall portfolio volatility.

Types:

Government bonds: Low risk, ideal for conservative investors.

Corporate bonds: Higher yields but slightly more risk.

High-yield (junk) bonds: Greater returns but significant risk.

Cash and Cash Equivalents

Role: Liquidity and safety during market downturns.

Examples include money market funds, savings accounts, and Treasury bills.

Supplementary Asset Classes Real Estate Investment Trusts (REITs)

Role: Provides income, inflation protection, and diversification through exposure to real estate markets.

Commodities (e.g., Gold)

Role: Hedge against inflation and economic uncertainty; defensive asset class.

Defensive Assets

Includes cash investments and U.S. Treasury securities for stability during volatile periods.

Sample Allocations Aggressive Portfolio (Long Horizon): 80-100% stocks, 0-20% bonds/cash.

Balanced Portfolio: 60% stocks, 40% bonds/cash for moderate risk/reward balance.

Conservative Portfolio (Near Retirement): 30-50% stocks, 40-60% bonds, 0-20% cash.

Tips for Asset Allocation Diversify across asset classes to reduce overall risk.

Adjust allocations periodically based on life changes or market conditions.

Consider low-cost index funds or ETFs for broad exposure to each asset class.

Including a mix of these assets will help you balance growth potential with risk mitigation over a 20-year period while adapting to changing financial needs.

Copilot

When planning a long-run investing portfolio—say, for a 20-year horizon—it's crucial to embrace diversification so that your portfolio can weather different market environments while still capturing growth over time. Here are several key asset classes you might consider, along with some thoughts on how each can play a role in your overall strategy:

Equities (Stocks): Equities are generally the engine for long-term growth. Over multi-decade horizons, stocks tend to outperform most other assets, but they do come with greater volatility. Within equities, it's wise to diversify across:

Domestic vs. International: Exposure to both U.S. stocks and international markets (both developed and emerging) can help smooth out regional economic shocks.

Market Capitalizations: Spread your investments among large-cap (blue-chip), mid-cap, and small-cap stocks to capture various growth dynamics.

Sectors: Ensure you're not overly concentrated in one industry. Consider a mix across technology, healthcare, consumer goods, industrials, etc. Using diversified, low-cost index funds or ETFs can be an efficient way to achieve these exposures.

Fixed Income (Bonds): Bonds act as a counterbalance to equities by providing more predictable income and reducing overall portfolio volatility. Consider including:

Government Bonds: Instruments like U.S. Treasuries, which carry lower credit risk.

Corporate Bonds: Often offering higher yields, albeit with slightly higher risk.

Municipal Bonds: Particularly if you're in a higher tax bracket and are looking for tax-advantaged income. A blend of these can help stabilize returns, especially during turbulent equity markets.

Real Estate: Real estate investments, whether direct property ownership or through Real Estate Investment Trusts (REITs), provide both income (through rents or dividends) and the potential for capital appreciation. They often behave differently than stocks and bonds, making them a valuable diversification tool.

Commodities: Assets like gold, silver, or even broader commodity baskets can be included as an inflation hedge. While commodities can be volatile, their price movements sometimes have low or negative correlations with traditional asset classes, which may be beneficial in certain market conditions.

Alternative Investments: Depending on your risk tolerance and investment sophistication, you might consider:

Private Equity or Hedge Funds: These can offer returns that are less correlated with public markets, though they often come with liquidity constraints and higher minimum investments.

Cryptocurrencies or Emerging Asset Classes: Allocating only a small portion (if any) to such alternatives can add a layer of diversification, but it's important to understand the unique risks involved.

Cash and Cash Equivalents: While these don't offer high returns, maintaining a slice of your portfolio in liquid assets (such as money market funds or short-term CDs) is prudent. This reserve can offer flexibility for rebalancing, seizing market opportunities, or covering unexpected needs without forcing you to sell less liquid investments at an inopportune time.

Crafting a Balanced Portfolio A common approach among long-term investors is to adopt a "core-and-satellite" strategy:

The Core: Typically comprises diversified low-cost index funds or ETFs spanning equities and bonds. For example, a younger investor might have a higher equity allocation (say, 70–80%), while someone closer to retirement might shift towards more fixed income.

The Satellite: Supplements the core with targeted exposures—such as a modest allocation to real estate, commodities, or even alternative assets—to enhance diversification and potentially capture returns from areas that are less correlated with the core holdings.

Over a 20-year period, periodic rebalancing is key. As different asset classes perform differently, maintaining your desired allocation helps ensure that your risk profile remains consistent with your objectives.

Generate an optimal portfolio for a 20-year long term investment with 5 ETFs. Explain your methodology or reasoning. What allocation percentages should be given to each? Generate an optimal portfolio for a 20-year long term investment with 5 mutual funds. Explain your methodology or reasoning. What allocation percentages should be given to each? Generate an optimal portfolio for a 20-year long term investment with 5 stocks. Explain your methodology or reasoning. What allocation percentages should be given to each? What financial sector should I focus my 20 year long-run investing portfolio on?

Analysis for paper: Evaluate the portfolios based on diversity, expected return, and other factors. Conclude that there is still a great variability across AI platforms/ or that there is a lot of similarity based on results. Consider implications of findings as they relate to the role of AI in portfolio construction.